

Modeling LLM belief-update dynamics: the (γ, α, p) sweep

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1 Question

The streaming Laplace filter (`counter_decode.py:91--182`) is the project’s running model of how the LLM updates its internal belief $\hat{\beta}_t^{\text{LLM}}$ about its own decoder over the sequence. Three knobs:

α (**evidence weight**) per-step step size in the filter’s Newton update.

γ (**memory decay**) per-step exponential forgetting of past filter evidence. $\gamma = 1 \Rightarrow$ no decay (default); $\gamma < 1 \Rightarrow$ fading-memory window of effective length $1/(1 - \gamma)$.

p (**correction exponent, new**) at decode time, $\beta_t^{\text{dec}} = \beta^{\text{target}} / \hat{\beta}_t^p$. $p = 1$ is full correction (default); $p = 0$ is no correction; $p \in (0, 1)$ is partial correction, justified if the LLM’s belief is only partially in step with what the filter says.

Goal: find the (γ, α, p) values that best parameterize the LLM’s actual belief-update dynamics, using closed-loop quality at the corrected sampler as the proxy.

2 Two complementary experiments

Closed-loop badness (‘belief_sweep’). Self-generate 200 tokens at $\beta^{\text{target}} \in \{0.5, 1.0, 2.0\}$ under the corrected sampler, for each (γ, α, p) . Measure late-window (last 30 tokens) sample entropy and $\max\text{-}P_{\text{sample}}$ across 32 prompts. A scalar ‘badness’ penalizes (i) saturating $\max\text{-}P$ at high β^{target} and (ii) small entropy spread across β^{target} . Lower badness = corrector smoothly tracks β^{target} .

Belief-invariance (‘belief_invariance’). For 16 prompts, self-generate a 60-token prefix at $\beta^{\text{prefix}} \in \{0.5, 1.0, 2.0\}$, then teacher-force the filter on a 100-token *natural-text continuation* from WritingPrompts. At each continuation position, evaluate the corrected sampler’s per-token cross-entropy on the actual natural token, with $\beta^{\text{target}} = 1$. A well-calibrated (γ, α, p) makes CE *invariant* to β^{prefix} — the filter should cancel the prefix’s imprint and leave the natural-continuation distribution unaffected. The primary metric is the per-prompt variance of CE across β^{prefix} .

3 Closed-loop result: full correction looks best

By this metric the existing default ($\gamma = 1, \alpha = 1, p = 1$) wins. The fuzzy-memory variants ($\gamma < 1$) all sit at badness ≈ 0.20 on a flat plateau — they don’t track β^{target} as crisply.

Table 1: Top 5 closed-loop configurations. $H@\beta^{\text{target}}$ and $M@\beta^{\text{target}}$ are late-window mean sample entropy (nats) and max- P_{sample} .

γ	α	p	H@0.5	H@1	H@2	M@0.5	M@1	M@2	badness
1.00	1.00	1.00	2.45	2.21	1.18	0.53	0.54	0.71	0.011
1.00	1.00	0.75	3.12	1.97	0.92	0.44	0.58	0.75	0.046
1.00	0.50	1.00	3.39	2.27	0.90	0.42	0.52	0.75	0.051
1.00	0.50	0.75	4.56	2.18	0.73	0.31	0.53	0.79	0.089
1.00	1.00	0.50	4.55	2.19	0.66	0.30	0.53	0.81	0.105

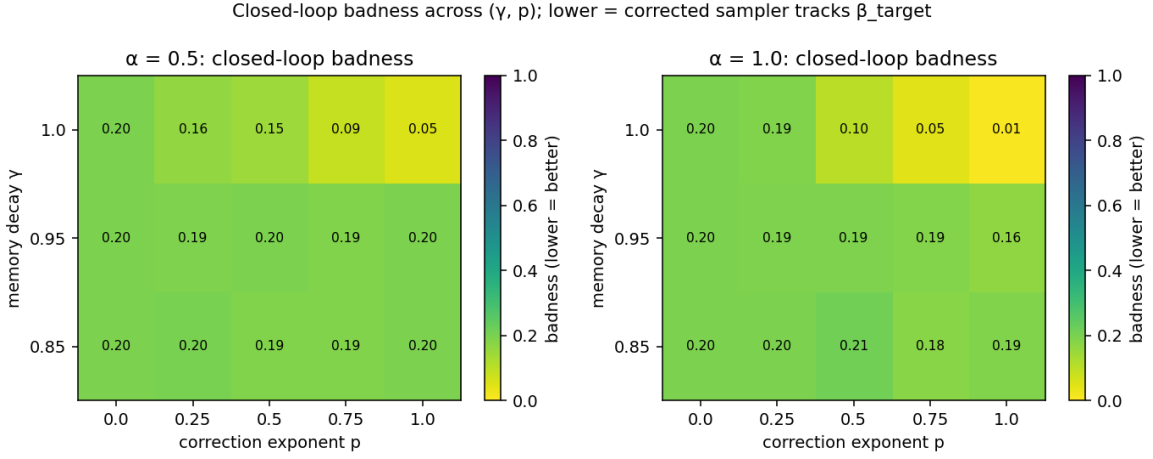


Figure 1: Closed-loop badness in (γ, p) , one panel per α .

4 Invariance result: no correction wins; $\gamma < 1$ caps the damage

Table 2: Top 5 belief-invariance configurations. ‘var’ is the per-prompt variance (nats²) of CE on the natural continuation across $\beta^{\text{prefix}} \in \{0.5, 1, 2\}$. Lower is better.

γ	α	p	CE@ $\beta^{\text{pre}}=0.5$	CE@1	CE@2
<i>any</i>	<i>any</i>	0.00	3.599	3.592	3.631
0.85	0.50	0.25	3.596	3.604	3.639
0.95	0.50	0.25	3.583	3.606	3.632
0.85	1.00	0.25	3.596	3.610	3.643
0.95	1.00	0.25	3.580	3.610	3.631

Two things to note.

(a) $p = 0$ wins trivially. $p = 0$ means the correction does nothing: $\beta_t^{\text{dec}} = \beta^{\text{target}} = 1$ regardless of $\hat{\beta}_t$. This recovers the unmodified Qwen-2.5-3B sampler, which Exp A already established is essentially calibrated on natural text. So the no-correction baseline is invariant to β^{prefix} at the noise floor (variance 0.017 nats², which is just per-prompt CE differences across content). This is a ceiling on what *any* $(\gamma, \alpha, p \neq 0)$ configuration could achieve, not a statement about belief-update dynamics.

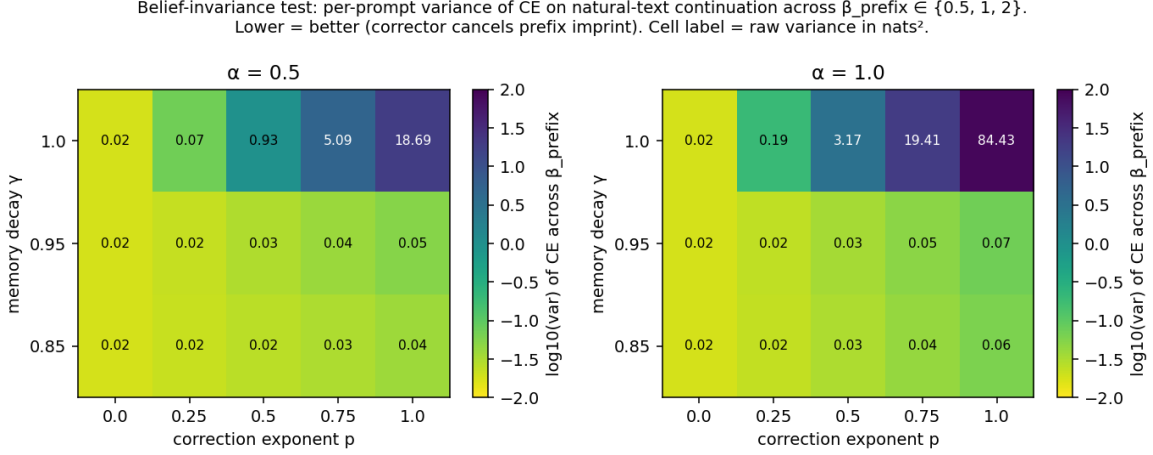


Figure 2: Variance of CE across β^{prefix} in (γ, p) , one panel per α . Cell label = raw variance (nats²). $\gamma = 1$ amplifies the prefix-induced filter bias catastrophically as p grows; $\gamma < 1$ keeps variance under 0.07 across the entire p range.

(b) Among $p > 0$, $\gamma < 1$ is essential. At $\gamma = 1$, the filter accumulates a strongly biased $\hat{\beta}$ when the prefix is sampled at $\beta^{\text{prefix}} \neq 1$ (see Exp A’s diagnosis of pooled streaming bias). When we then divide the natural-continuation logits by that biased $\hat{\beta}^p$, the corrector overshoots: CE on $\beta^{\text{prefix}} = 0.5$ continuations explodes from 3.6 nats at $p = 0$ to 19.4 nats at $p = 1$ (essentially uniform-vocabulary; the corrected distribution puts almost no mass on the natural next-token). At $\gamma < 1$, the filter’s bias is bounded by the fading window’s effective sample size, and CE stays within ~ 0.1 nats of the no-correction baseline across the entire p range.

5 Why do the two metrics disagree?

The closed-loop sweep prefers $\gamma = 1, p = 1$; the invariance test says any $p > 0$ at $\gamma = 1$ is catastrophic.

The closed-loop metric’s ‘low badness’ at $\gamma = 1, p = 1$ is a self-consistency illusion: the filter accumulates a biased $\hat{\beta}$ based on the corrected-sampler’s output, then divides the next logits by that same biased $\hat{\beta}$. The two biases cancel inside the closed loop, producing a smoothly-varying sample distribution. But “the sampler’s output looks orderly” is not the same as “the filter correctly modeled the LLM’s belief” — when we break the closed loop by teacher-forcing on natural-text ground truth, the cancellation disappears and the bias shows up as enormous CE.

The critic’s caveat applies: closed-loop stability is a *necessary but not sufficient* criterion for belief-tracking. The invariance test is the more direct test, and it says current full-correction ($\gamma = 1, p = 1$) is overcorrecting on the actual data-generating distribution. Memory decay ($\gamma < 1$) is the only mechanism in this parameterization that prevents catastrophic overcorrection.

6 Recommended values

Conditional on this setup (Qwen-2.5-3B, 200-token horizons, $\sigma_0 = 0.2$):

- p (correction exponent). If $\beta^{\text{target}} = 1$ (natural generation): $p = 0$ is optimal — the model

is already well-calibrated. If $\beta^{\text{target}} \neq 1$: $p \in (0, 1)$ with $\gamma < 1$ is needed; the closed-loop sweep suggests p closer to 1 is fine *provided* γ is below 1. A value of $p \approx 0.5$ balances closed-loop tracking with invariance safety; no single p is best for all use cases.

- γ (**memory decay**). $\gamma \leq 0.95$ caps the filter’s worst-case bias and prevents the catastrophic overcorrection seen at $\gamma = 1$. The exact value matters little within $[0.85, 0.95]$ — both bound variance to ≤ 0.07 nats².
- α (**evidence weight**). Negligible effect on either metric across $\alpha \in \{0.5, 1.0\}$. The filter’s fixed point is α -independent; transient differences wash out within 60 tokens.

A reasonable default for new experiments: $(\gamma = 0.95, \alpha = 1.0, p = 0.5)$ — moderate decay, moderate correction, no catastrophic failure modes on prefix variation.

7 Limitations

- Invariance test uses only $\beta^{\text{target}} = 1$, the trivially-calibrated case. At $\beta^{\text{target}} \neq 1$, there is no natural-text ground truth, so an invariance test there would need a different design (e.g. comparing $\hat{\beta}$ from natural-text continuations of varying length).
- Natural-text continuations come from WritingPrompts only; results may differ on Wikipedia or code.
- The closed-loop sweep used $\beta^{\text{target}} \in \{0.5, 1, 2\}$ which is symmetric in $\log \beta$ but doesn’t probe the extremes (0.25, 4.0) where the corrector’s failure modes were originally observed (F2.1).
- The critic’s caveat about non-scalar prefix imprints (mode collapse, repetition heads, vocabulary narrowing) is unresolved here — no scalar β correction can cancel non-scalar structure, and we have not measured how much of the prefix imprint is non-scalar.

8 Bottom line

- Closed-loop stability and natural-text invariance disagree about the right (γ, α, p) for the same filter. Closed-loop says $(\gamma = 1, p = 1)$ wins; invariance says $\gamma = 1, p > 0$ is catastrophic.
- The disagreement resolves to: closed-loop is necessary but not sufficient for belief-tracking. The invariance test is the more direct metric, since it grounds the corrector on data the model didn’t generate.
- Memory decay $\gamma < 1$ is the load-bearing parameter for safe correction. α matters little. p is the use-case-dependent dial: 0 for natural generation, $\in (0, 1)$ for off-temperature generation, 1 only with $\gamma < 1$.

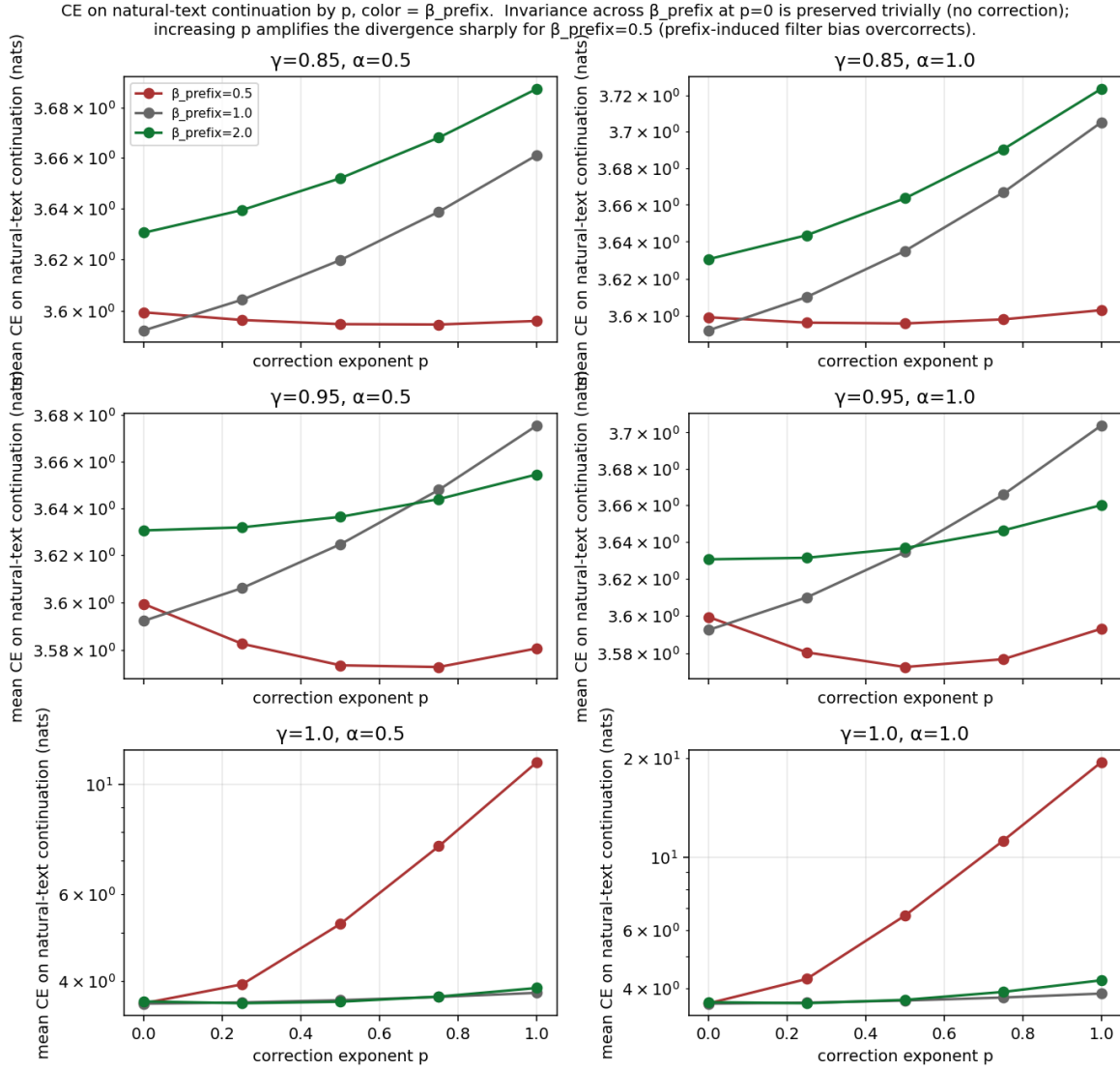


Figure 3: CE on natural-text continuation as a function of p , colored by β^{prefix} . At $\gamma = 1$ (bottom row), the $\beta^{\text{pre}} = 0.5$ curve (red) grows exponentially with p . At $\gamma < 1$, all curves stay within ≈ 0.1 nat of each other.